



DATA ANALYTICS

Customer Shopping Behaviour Analysis

Analyzing 3,900 customer transactions to uncover actionable insights on revenue, loyalty, discounts, and purchasing patterns using Python and SQL.

Dataset Overview

3,900

Total Transactions

Customer shopping records analyzed

19

Data Columns

Features after cleaning and engineering

37

Missing Values

Filled using category-based median

Numerical Features

- Age and purchase amount
- Review ratings
- Previous purchases

Categorical Features

- Gender and product category
- Shipping type
- Subscription status

Data Cleaning & Feature Engineering

01

Missing Value Treatment

Filled 37 missing review ratings using median values within each product category for accuracy.

03

Age Segmentation

Created four customer groups using quartiles: Young, Adult, Middle Age, and Senior.

02

Column Standardization

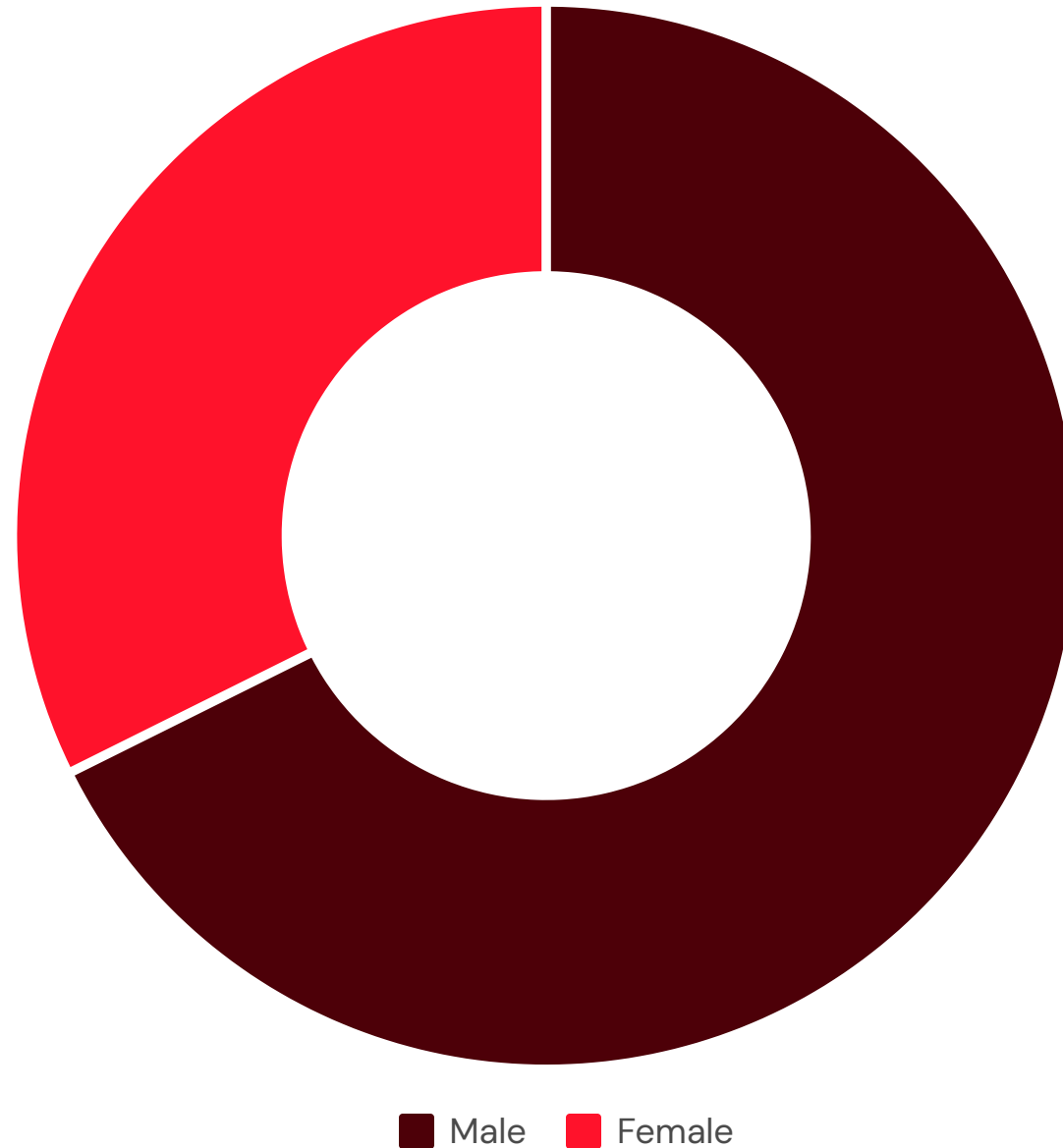
Converted all column names to lowercase and replaced spaces with underscores for SQL compatibility.

04

Purchase Frequency Mapping

Converted frequency labels to numeric days (Weekly=7, Monthly=30, Quarterly=90, Annually=365).

Revenue by Gender



Male customers contribute 68% of total revenue, representing the dominant purchasing segment. This insight enables targeted marketing strategies and product positioning.

Discount Strategy Insights

High-Value Discount Users

Customers using discounts while spending **above average**:

- Customer #2: \$64
- Customer #3: \$73
- Customer #4: \$90
- Customer #7: \$85
- Customer #9: \$97



❏ **Key Finding:** Discounts don't always reduce revenue—strategic customers use them while maintaining high purchase values.

Product Performance Analysis

1

Gloves

Rating: **3.9**

Highest customer satisfaction

2

Sandals

Rating: **3.8**

Strong performance

3

Boots

Rating: **3.8**

High brand trust

4

Hat

Rating: **3.8**

Consistent quality

5

Skirt

Rating: **3.8**

Customer favorite

Shipping & Subscription Impact

Express Shipping

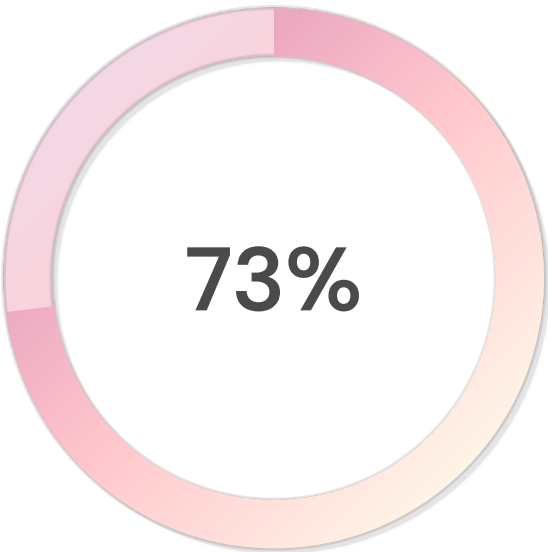
Average Spend: **\$60.48**

Associated with higher-value purchases and premium customer segment.

Standard Shipping

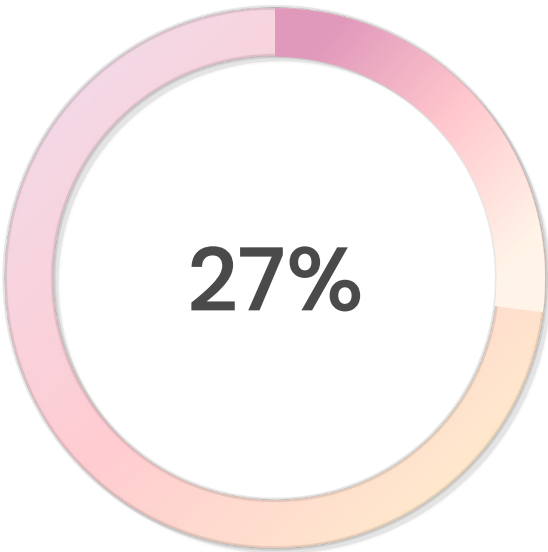
Average Spend: **\$58.46**

Preferred by cost-conscious shoppers seeking value.



Non-Subscribers

2,847 customers generating \$170,436 total revenue

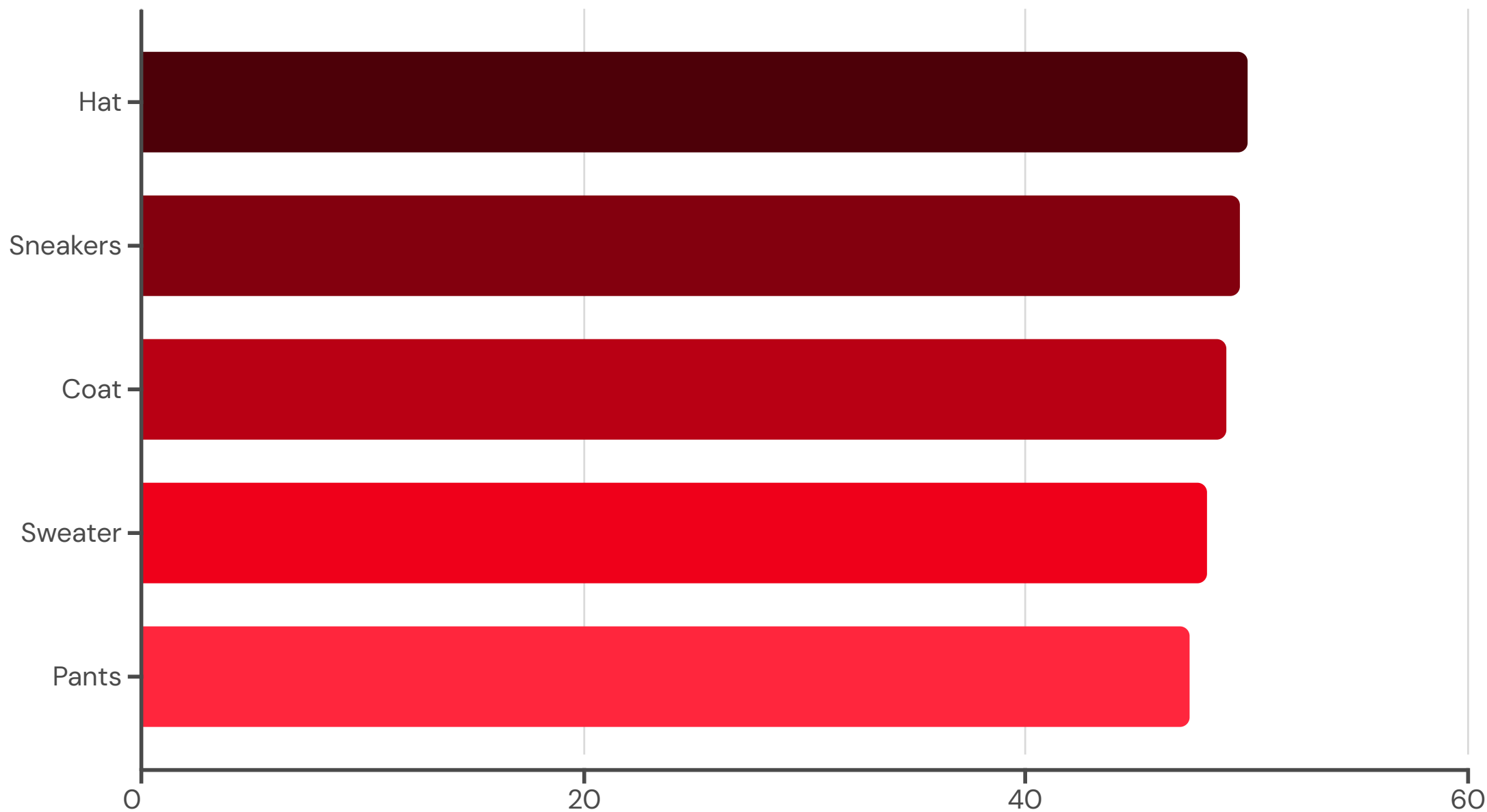


Subscribers

1,053 customers generating \$62,645 with higher lifetime value potential

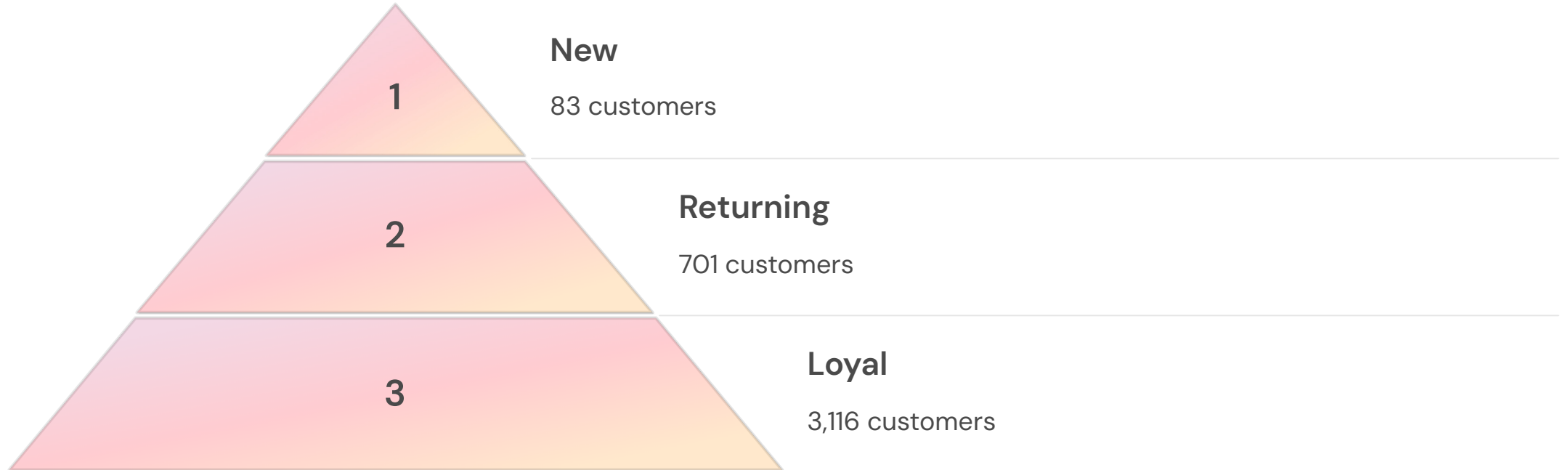
Discount-Driven Products

Products with highest percentage of discounted purchases reveal promotion-sensitive categories:



Nearly half of these products are sold with discounts, indicating strong price sensitivity and promotional effectiveness.

Customer Loyalty Segmentation



80% of customers are loyal with more than 10 previous purchases, forming a strong foundation for long-term revenue growth.

Among repeat buyers (5+ purchases), **958 are subscribers** versus 2,518 non-subscribers, showing subscription correlates with trust and engagement.

Key Business Takeaways



Loyalty Drives Revenue

Subscriptions and repeat purchases strongly influence customer lifetime value and total revenue generation.



Strategic Discounting

Discounts coexist with high spending behavior—premium customers use promotions without reducing purchase value.



Age Group Targeting

Young customers generate \$62,143 in revenue, representing the most profitable segment for targeted marketing campaigns.



Quality Indicators

Product ratings serve as useful indicators for premium positioning and promotional strategy development.